Batterjee Medical College
For Science and Technology

BMC

Acid Base Disorders

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Learning Objectives

Knowledge:

•Describe the gross structure and functional anatomy of the kidney, including nephron segments.
•Explain renal functions such as waste excretion, fluid volume control, plasma pH regulation via HCO_3^-
• handling, and hormonal roles.
•Define acidosis, alkalosis, acidemia, alkalemia, and the physiological buffer mechanisms.
•Identify normal arterial blood gas (ABG) reference ranges (pH, $PaCO_2$, PaO_2, HCO_3^-)
•Explain the underlying mechanisms, common causes, and clinical manifestations for respiratory acidosis, respiratory alkalosis, metabolic acidosis and metabolic alkalosis.
•Recognize the clinical picture of metabolic acidosis, respiratory acidosis, metabolic alkalosis and respiratory alkalosis.
•Describe evidence-based treatment strategies for each acid-base disorder, including underlying cause management, oxygen therapy, mechanical ventilation, bicarbonate, and fomepizole/hemodialysis.

Skills:

•Systematically analyze blood gas parameters to differentiate between simple, uncompensated, and compensated acid-base disorders.
•Calculate and interpret the serum Anion Gap to categorize metabolic acidosis.
•Utilize tools like the "Tic-Tac-Toe" grid and the SMORE mnemonic to identify primary disturbances.
•Apply compensation formulas to determine if appropriate physiological compensation is present or if a mixed disorder exists.
•Formulate initial evidence-based management and therapeutic plans based on patient presentation and diagnostic laboratory results.

Values:

•Demonstrate professional behavior by adhering to institutional training, clinical rules, and established medical regulations.
•Appreciate the critical importance of prompt, accurate acid-base assessment in managing acutely ill patients safely and ethically.

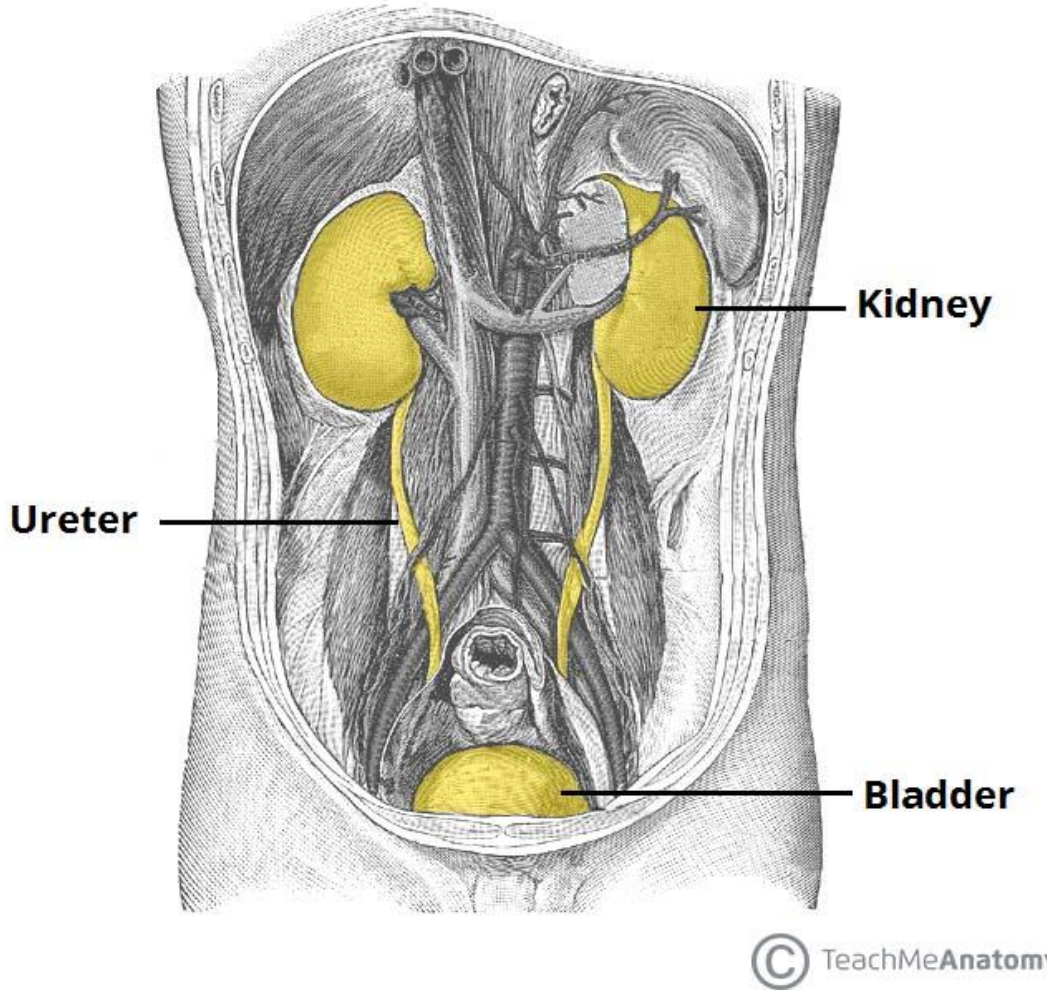
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- Respiratory Acidosis
- Respiratory Alkalosis
- Metabolic Acidosis
- Metabolic Alkalosis
- Patient Approach

Introduction

Kidneys Lie retroperitoneal, on both side of vertebral column at level of T12 to L3 although the right kidney is often situated slightly lower due to the presence of the liver.

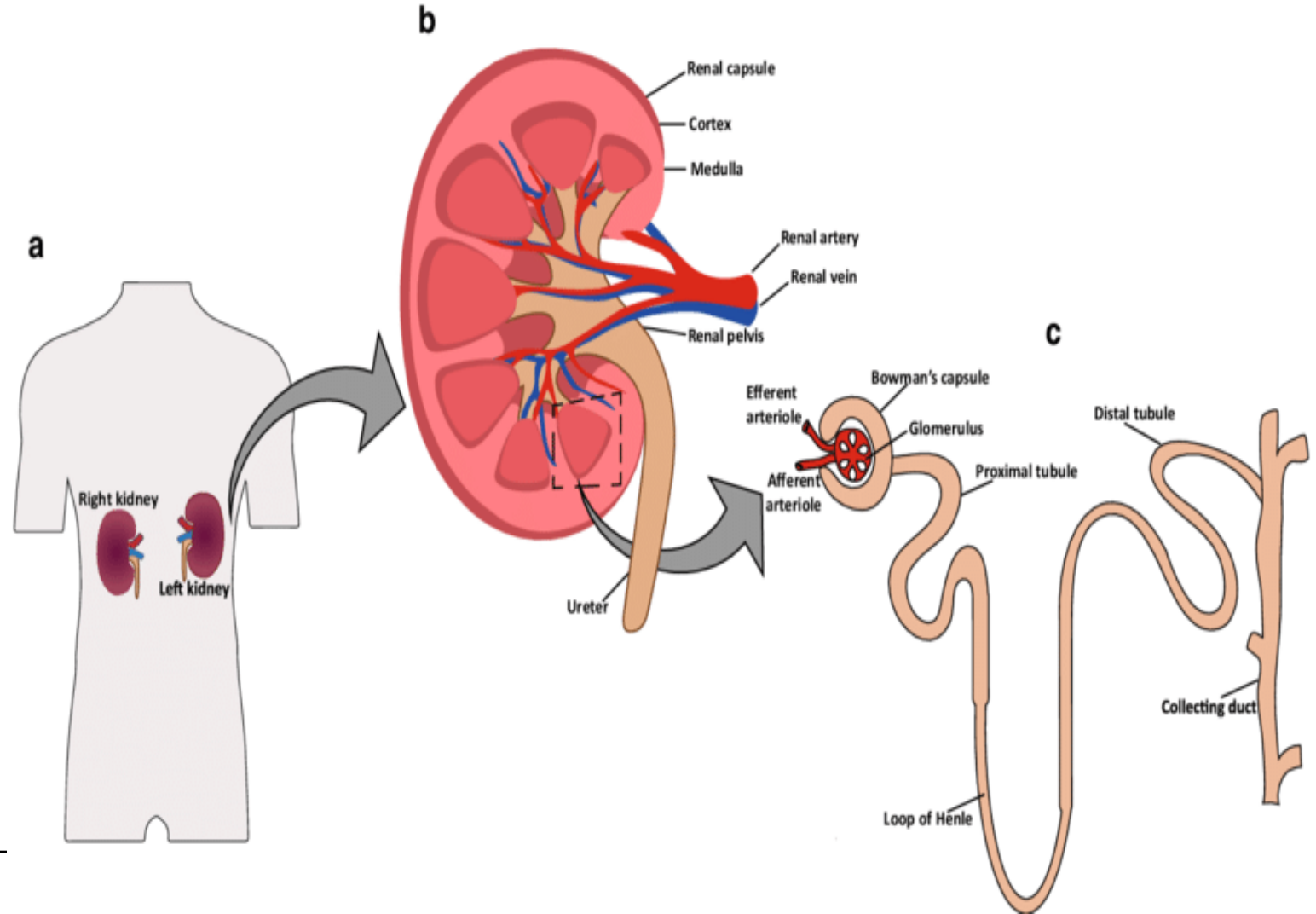
On bisection, a simple gross structure is evident with an outer cortex and a more central medulla.



Introduction

The functional unit of the kidney is the nephron

Each nephron is made up of glomerulus, proximal tubules, loop of henel, distal tubules and collecting duct.



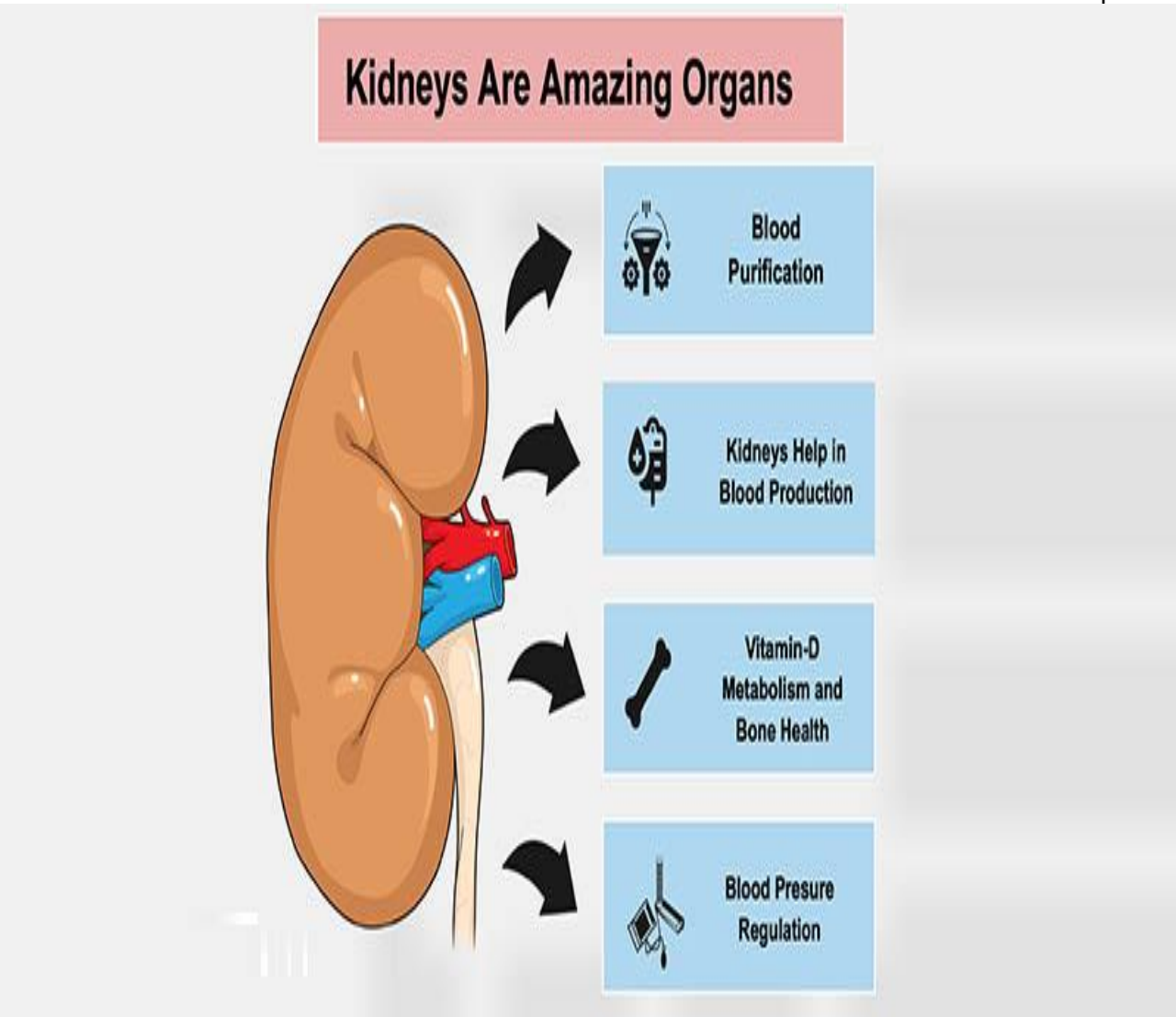
Function of the kidney

Excretory: excretion of waste products

Regulatory: regulates body fluid volume & its contents

Endocrine: production of erythropoietin, renin, angiotensin & prostaglandins

Metabolic: metabolism of vitamin D, small molecular weight proteins like insulin.



The Kidney Regulates Extracellular Fluid Volume By Regulating Its Sodium Content

Filtration & reabsorption of sodium.

A number of the steps that regulate sodium reabsorption can be counteracted by atrial natriuretic peptide (ANP), which is released from the atria in response to volume overload.

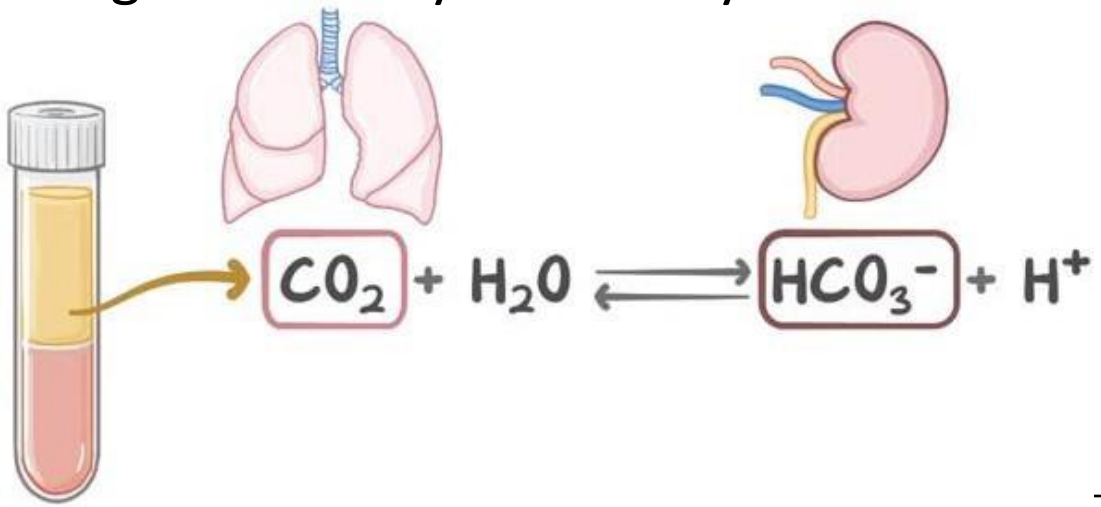
It Control the osmolarity of the body fluids through controlling of intake and excretion, whereby the kidney can vary the osmolarity of urine.

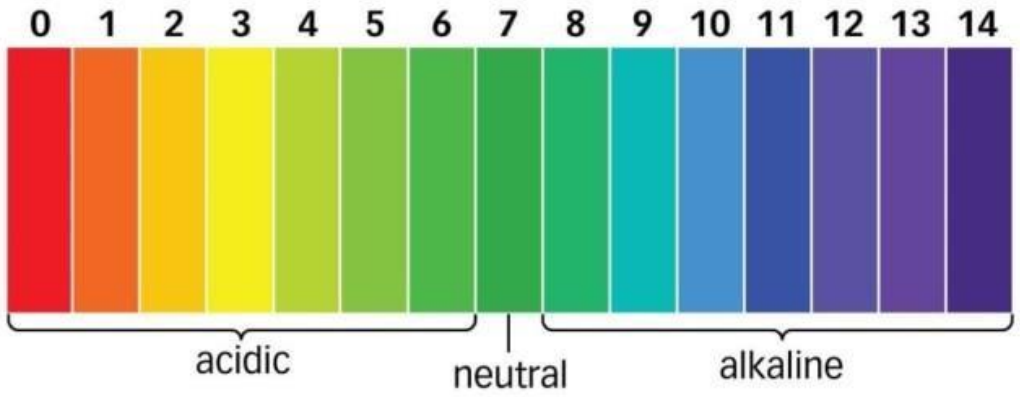
Antidiuretic hormone (ADH), also called vasopressin, is a critical component of water reclamation.

ADH is released by changes in osmolarity of less than 1%.

The Kidney Regulates Plasma PH By Regulating HCO3 Content

- Blood pH refers to the level of H+ ions
- Acid-base balance is concerned with maintaining a normal hydrogen ion concentration in the body
- This balance is achieved by buffers utilization by respiratory and renal systems
- HCO3 – is the most important buffer not only because it is consumed by metabolic acid, thereby producing CO2 that can be exhaled, but also because it can be regenerated by the kidney.





Acid base disorders

- Acid-base disorders are characterized by changes in the concentration of hydrogen ions (H+) in the body.

•Acid-base processes

- Acidosis: the processes by which H^+ concentration is increased
- Alkalosis: the processes by which H^+ concentration is decreased

•Blood pH abnormalities

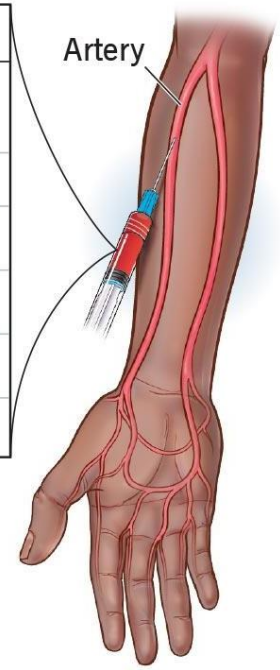
- Acidemia: abnormally low blood pH (pH < 7.35)
- Alkalemia: abnormally high blood pH (pH > 7.45)

- If Compensation occurs, acidosis and/or alkalosis may be present without acidemia or alkalemia.

- assessment of acid base abnormalities is typically done using arterial blood gases (ABG) or venous blood gases (VBG)

Arterial Blood Gas (ABG)

ABG	Normal range
O_2CT	15-23% per 100 mL of blood
pH	7.35-7.45
$PaCO_2$	35-45 mmHg
PaO_2	80-100 mmHg
HCO_3	22-26 mEq/L
O_2Sat	95-100%



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Respiratory Acidosis

Definition: ↑ PCO_2 & ↓ pH

Mechanism: Alveolar hypoventilation → CO_2 retention → Increases in PaCO2 → pH decreases

Common Causes:

Central (CNS)	Drugs, strokes, trauma, hemorrhage
Airway	Obstruction, asthma
Parenchyma	Emphysema, pneumoconiosis, bronchitis, adult respiratory distress syndrome, barotrauma, pneumonia
Neuromuscular	Poliomyelitis, kyphoscoliosis, myasthenia, muscular dystrophies
Miscellaneous	Obesity, hypoventilation, amyotrophic lateral sclerosis

Respiratory Acidosis

Clinical Features:

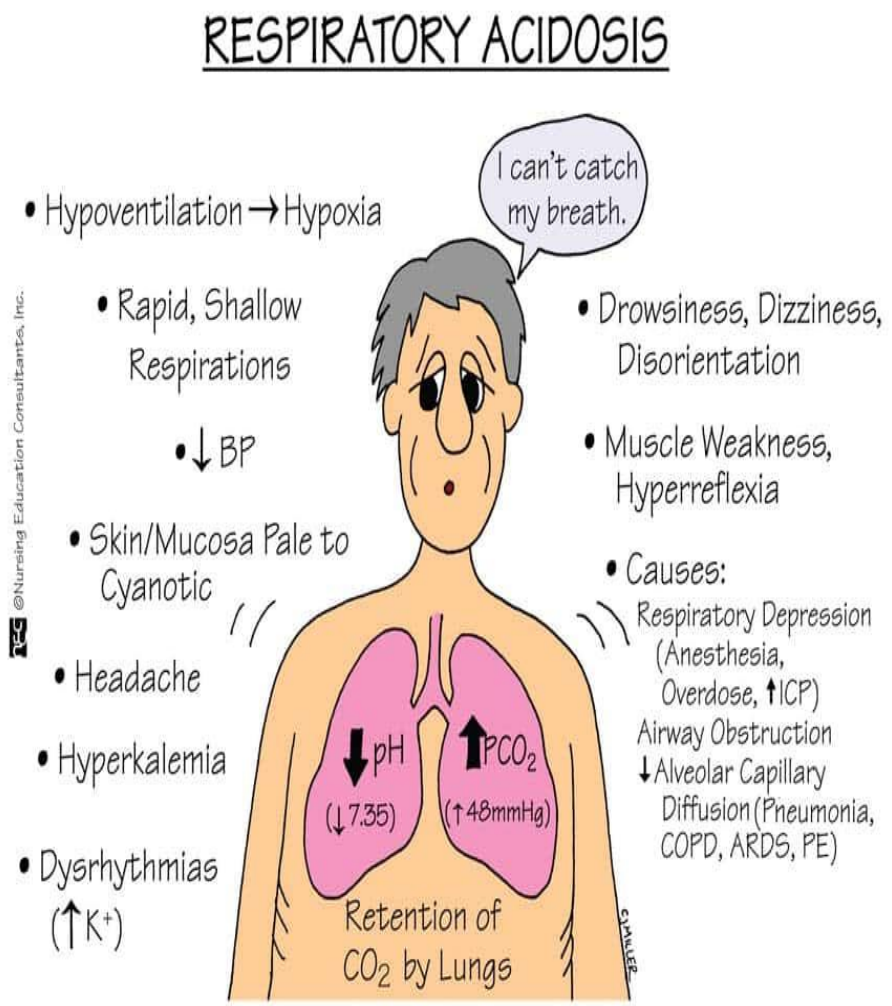
- Somnolence, myoclonus with asterixis, headaches. (picture of respiratory failure)

Compensation:

- HCO_3 will increase in blood (↑absorption by PCT, ↑ excretion of H^+ by DCT)
- Acute: HCO_3 ↑ by 1 mEq/l for every 10 mmHg ↑ in $PaCO_2$
- Chronic: HCO_3 ↑ by 3.5 mEq/l for every 10 mmHg ↑ in $PaCO_2$

Treatment:

- Treat underlying cause
- Give supplemental oxygen if PaO_2 is low (Be careful with COPD patients)
- Intubation and mechanical ventilation might be required for severe acidosis (pH < 7), Mental deterioration, Impending respiratory fatigue



Respiratory Alkalosis

Definition: ↓ PCO_2 & ↑ pH

Mechanism: ↑ Respiratory rate and/or tidal volume → alveolar hyperventilation → CO_2 washout

Common Causes:

CNS stimulation	Pain, Anxiety, psychosis, Fever, Cerebrovascular accident, Meningitis, encephalitis, Tumor, Trauma
Hypoxemia	High altitude, Pneumonia, pulmonary edema, severe anemia , PE
Drugs or hormones	Pregnancy, Salicylates (early)
Miscellaneous	Septicemia, hepatic failure, mechanical hyperventilation, heat exposure

Respiratory Alkalosis

Clinical Features:

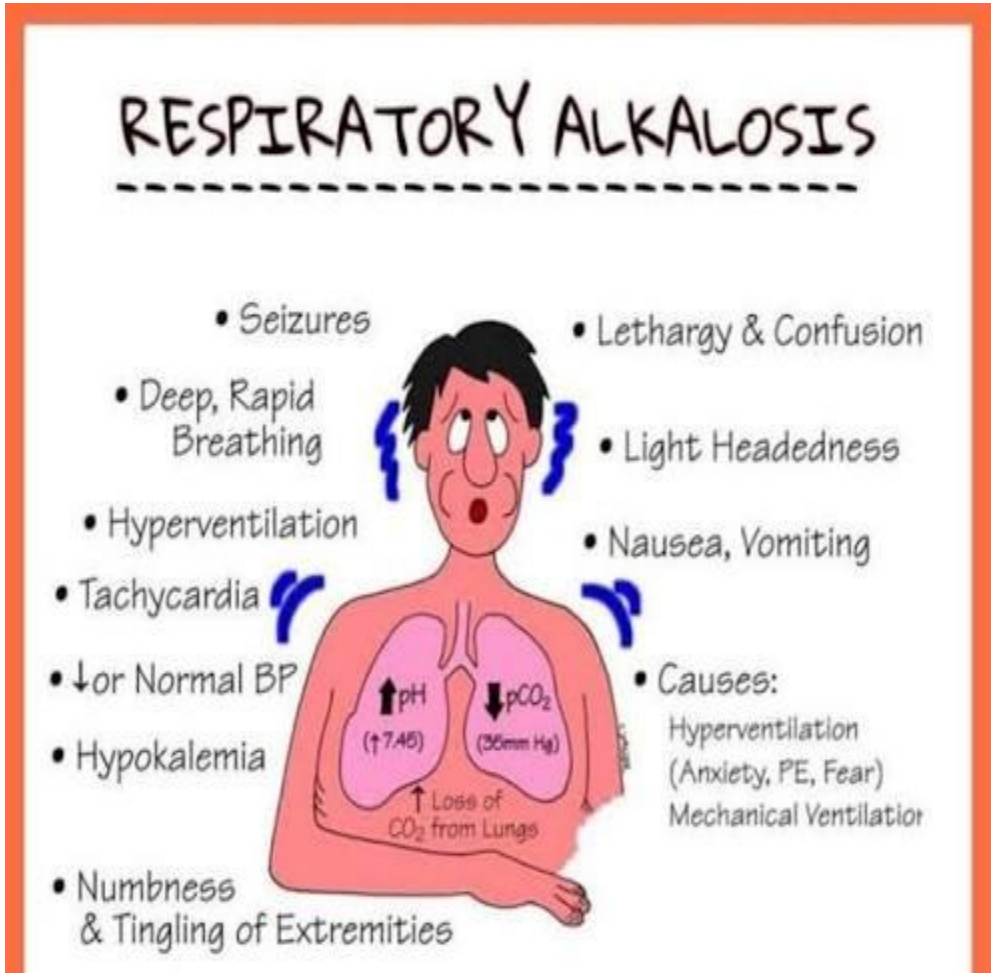
- lightheadedness, dizziness, anxiety, paresthesia, and perioral numbness
- Confusion and chest discomfort

Compensation:

- HCO_3 will decrease in blood
- Acute: HCO_3 ↓ by 2 mEq/l for every 10 mmHg ↓ in $PaCO_2$
- Chronic: HCO_3 ↓ by 4 mEq/l for every 10 mmHg ↓ in $PaCO_2$

Treatment

- Treat the underlying cause
- Sometimes this does not need to be treated (e.g., in the case of pregnancy)
- Breath into paper bag to recycle the exhaled CO2 (usually for those who have anxiety)



Metabolic Acidosis

Definition: ↓ HCO_3 & ↓ pH

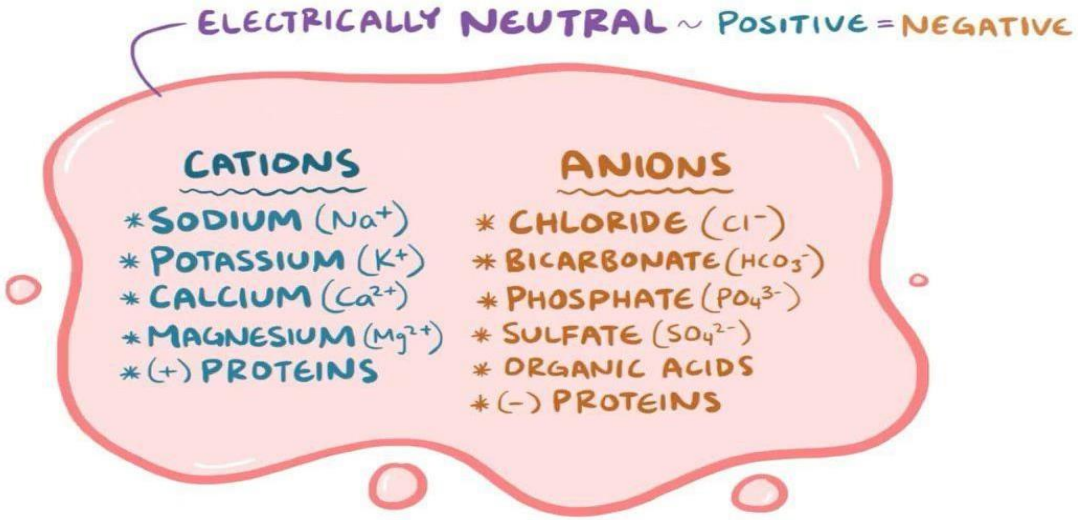
Mechanism: ↑ Production/ingestion of H^+ or loss of HCO_3

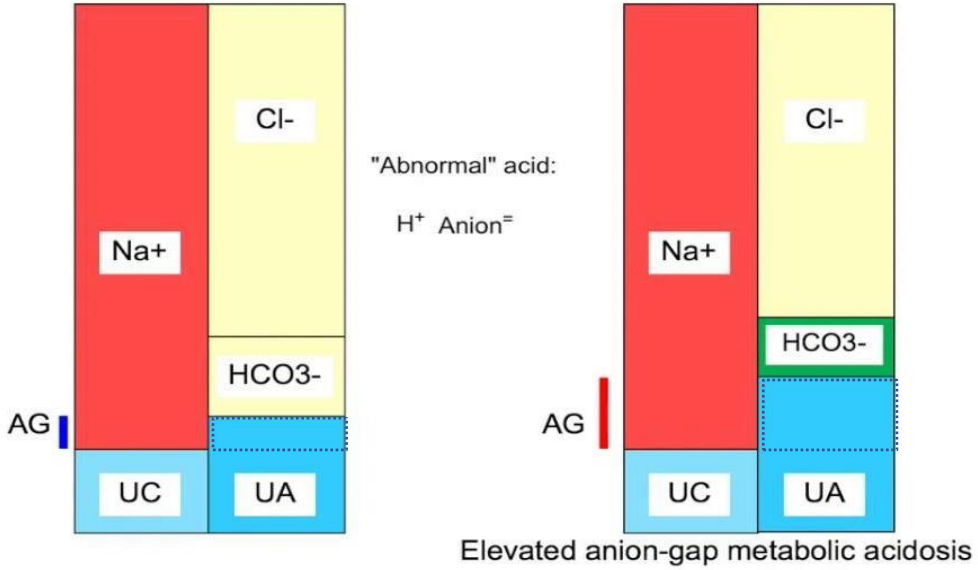
Common Causes:

High Anion Gap	**M**ethanol, **U**remia, **D**KA , **P**ropylene glycol, **I**NH, **L**actic acidosis, **E**thanol/**E**thylene Glycol , **S**alicylates toxicity (late)
Normal Anion Gap	**S**pironolactone, **H**yperalimentation, **A**cetazolamide/**A**ddison's disease, **R**enal tubular acidosis, **D**iarrhea, **S**aline infusion

Metabolic Acidosis

- The anion gap (AG) is the difference between primary measured cations and the primary measured anions
- The AG is calculated as follows: **AG = Na^+ – (Cl^- + $HCO3^-$)**
- The "normal" value for the AG ranges from 6 to 12 mmol/L
- Another Formula: ([Na^+] + [K^+]) - ([Cl^-] + [HCO_3^-]), with a normal range of 10–16 mEq/L.





Elevated anion-gap metabolic acidosis

Metabolic Acidosis

Clinical Features:

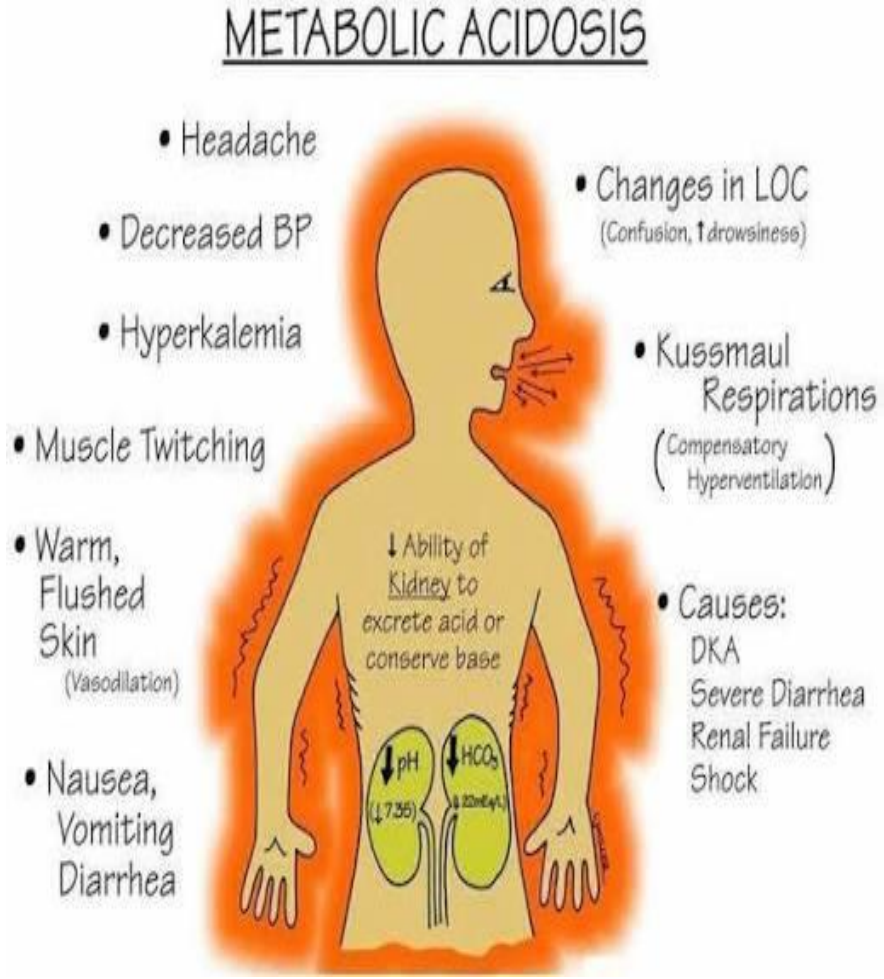
- Hyperventilation (Kussmaul respiration)
- Decreased in Cardiac output and tissue perfusion

Compensation:

- Hyperventilation → decrease $PaCO_2$ immediately
- **Winter's formula:** predicted $PaCO_2 = (1.5 \times [HCO_3] + 8) \pm 2$

Treatment:

- Treat the underlying cause
- Sodium bicarbonate is sometimes used in severe acidosis
- Mechanical ventilation might be needed if the patient is fatigued
- For ethylene glycol and methanol ingestion, treat with fomepizole and hemodialysis



Metabolic Alkalosis

Definition: ↑ HCO_3 & ↑ pH

Mechanism: Loss of H^+ or ↑ production/ingestion of HCO_3

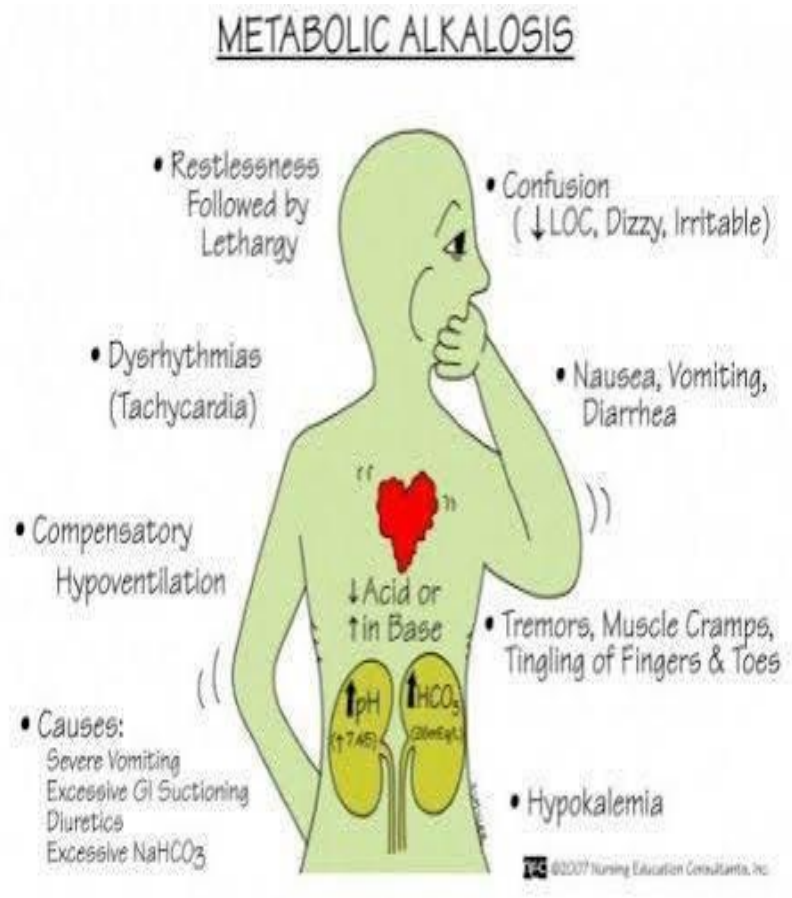
Common Causes:

• **Chloride-responsive metabolic alkalosis (urine chloride < 25 mEq/L)**	• Hypovolemia, e.g., contraction alkalosis • Renal losses (diuretics) • Cystic fibrosis • Dietary chloride deficiency
• **Chloride-resistant metabolic alkalosis (urine chloride > 40 mEq/L)**	• Loop or thiazide diuretics • Hyperaldosteronism • Cushing syndrome • High alkali load (e.g., due to antacid use, alkalization therapy)

Metabolic Alkalosis

Clinical Features:

- There are no characteristic signs and symptoms
- Muscle weakness, cramps, spasms, tremors, fatigue, irritability, and changes in mental status like confusion or seizures
- The patient's medical history is most helpful (look for vomiting, gastric drainage, diuretic therapy, and so on)



Compensation:

- Hypoventilation → increases $PaCO_2$ immediately
- $PaCO_2 = [0.7 \times (HCO_3 - 24)] + 40 \pm 2$

The first step if metabolic alkalosis is diagnosed is to measure urinary chloride

Metabolic Alkalosis

Treatment:

- **Chloride-responsive metabolic alkalosis**
  - Start isotonic saline to increase urinary bicarbonate excretion and correct extracellular volume loss

- **Chloride-resistant metabolic alkalosis**
  - Consider bicarbonate excess as a potential cause and administer acetazolamide.

Patient Approach

- PH and pCO_2 are the most important lab values to assess the acid-base status in any patient

- **Evaluate blood pH**:
  - pH < 7.35 (acidemia): Primary disorder is an acidosis
  - pH > 7.45 (alkalemia): Primary disorder is an alkalosis
  - pH 7.35-7.45 (normal) → Normal or mixed acidosis and alkalosis



Patient approach

Evaluate pCO2 (reference range 35-45 mm Hg)

- pH and pCO2 change in the **opposite direction**: respiratory disorder
- ↓pH and ↑pCO2: respiratory acidosis
- ↑pH and ↓pCO2: respiratory alkalosis

- pH and pCO2 change in the **same direction**: metabolic disorder
- ↓pH and ↓pCO2: metabolic acidosis
- ↑pH and ↑pCO2: metabolic alkalosis

SMORE: change of pCO_2 in the **S**ame direction as pH → **M**etabolic disorder; change of pCO_2 in the **O**pposite direction of pH → **RE**spiratory disorder

Patient Approach

- **Evaluate HCO_3^-** (reference range: 22–26 mEq/L):
  - High: metabolic alkalosis or compensated respiratory acidosis
  - Normal: uncompensated respiratory disorders
  - Low: metabolic acidosis or compensated respiratory alkalosis
- **Evaluate pO_2**:
  - High: hyperoxemia
  - Low: hypoxemia
- **Determine chronicity and compensation using the formulas**
- **Calculate the AG if applicable**

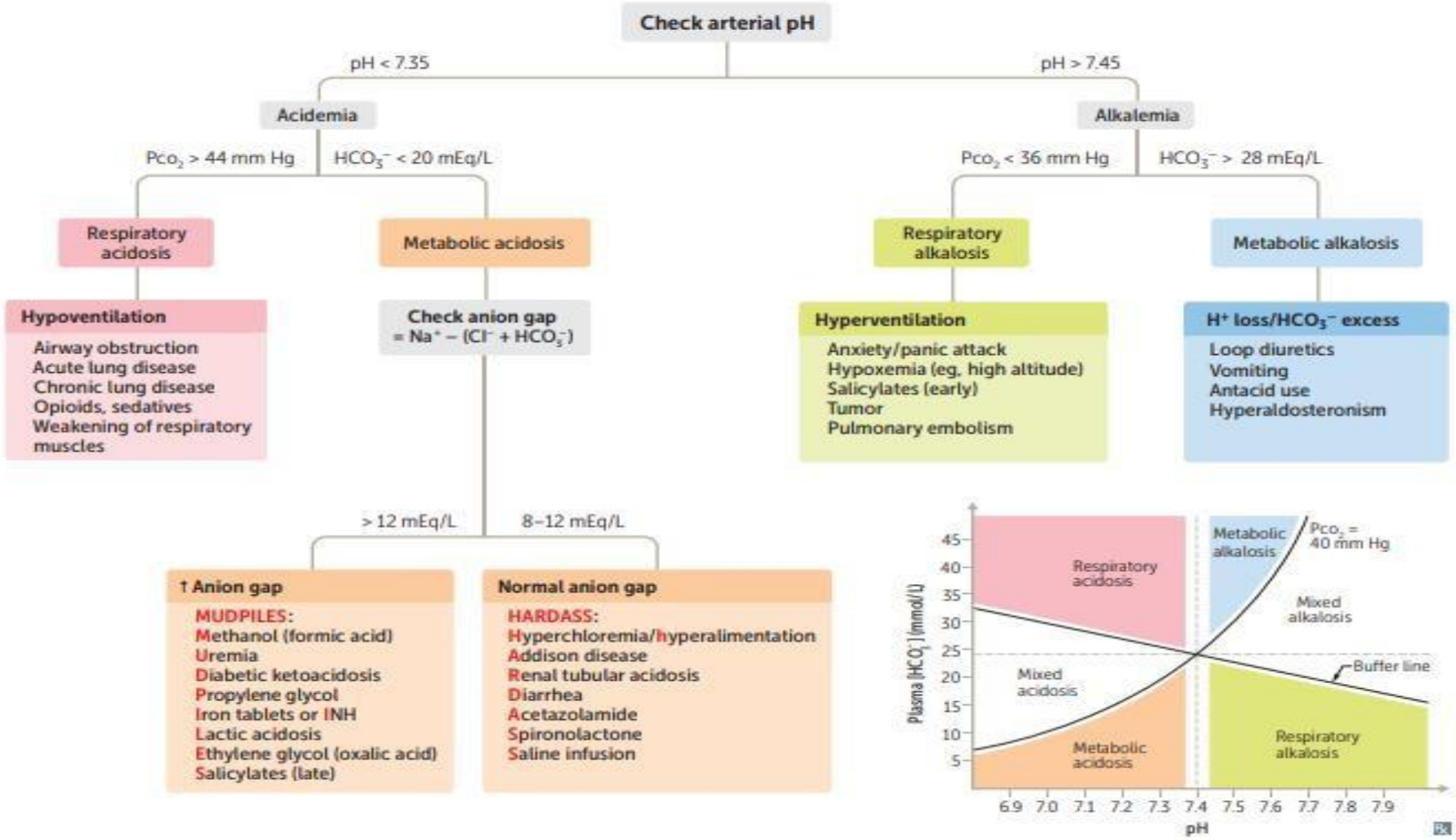
The anion gap should be checked even in the absence of an acidemia as a metabolic acidosis may be hiding!

Patient Approach

Calculation of compensatory response		
Primary acid-base disturbance		Expected compensation [1][9]
Metabolic acidosis		• Winter formula: expected PCO_2 (mm Hg) = (1.5 × HCO_3^-) + 8 ± 2
Metabolic alkalosis		• Expected PCO_2 (mm Hg) = [0.7 × (HCO_3^- - 24)] + 40 ± 2
Respiratory acidosis	Acute	• Expected HCO_3^- (mEq/L) = 24 + [0.1 × (PCO_2 - 40)]
	Chronic	• Expected HCO_3^- (mEq/L) = 24 + [0.35 × (PCO_2 - 40)]
Respiratory alkalosis	Acute	• Expected HCO_3^- (mEq/L) = 24 - [0.2 × (40 - PCO_2)]
	Chronic	• Expected HCO_3^- (mEq/L) = 24 - [0.4 × (40 - PCO_2)]

If compensation is inadequate, then there may be another process.
(i.e. if a patient's expected pCO2 is lower than the calculated compensation, there is a concomitant respiratory alkalosis).
The body will **never overcompensate** for a primary process.

Summary


Check arterial pH
pH < 7.35
pH > 7.45
Acidemia
Alkalemia
Pco2 > 44 mm Hg
HCO3− < 20 mEq/L
Pco2 < 36 mm Hg
HCO3− > 28 mEq/L
Respiratory acidosis
Metabolic acidosis
Respiratory alkalosis
Metabolic alkalosis
Hypoventilation
Airway obstruction
Acute lung disease
Chronic lung disease
Opioids, sedatives
Weakening of respiratory muscles
Check anion gap
= Na+ − (Cl− + HCO3−)
Hyperventilation
Anxiety/panic attack
Hypoxemia (eg, high altitude)
Salicylates (early)
Tumor
Pulmonary embolism
H+ loss/HCO3− excess
Loop diuretics
Vomiting
Antacid use
Hyperaldosteronism
> 12 mEq/L
8–12 mEq/L
↑ Anion gap
MUDPILES:
Methanol (formic acid)
Uremia
Diabetic ketoacidosis
Propylene glycol
Iron tablets or INH
Lactic acidosis
Ethylene glycol (oxalic acid)
Salicylates (late)
Normal anion gap
HARDASS:
Hyperchloremia/hyperalimentation
Addison disease
Renal tubular acidosis
Diarrhea
Acetazolamide
Spironolactone
Saline infusion
Plasma [HCO3−] (mmol/L)
45
40
35
30
25
20
15
10
5
Respiratory acidosis
Metabolic alkalosis
Pco2 = 40 mm Hg
Mixed alkalosis
Buffer line
Mixed acidosis
Metabolic acidosis
Respiratory alkalosis
6.9
7.0
7.1
7.2
7.3
7.4
7.5
7.6
7.7
7.8
7.9
pH

Case 1:

- pH 7.24 / pCO2 22 / pO2 92
- Na 138 / Cl 102 / HCO3 15

What's the most likely cause of the acid base disorder?

Case 1:

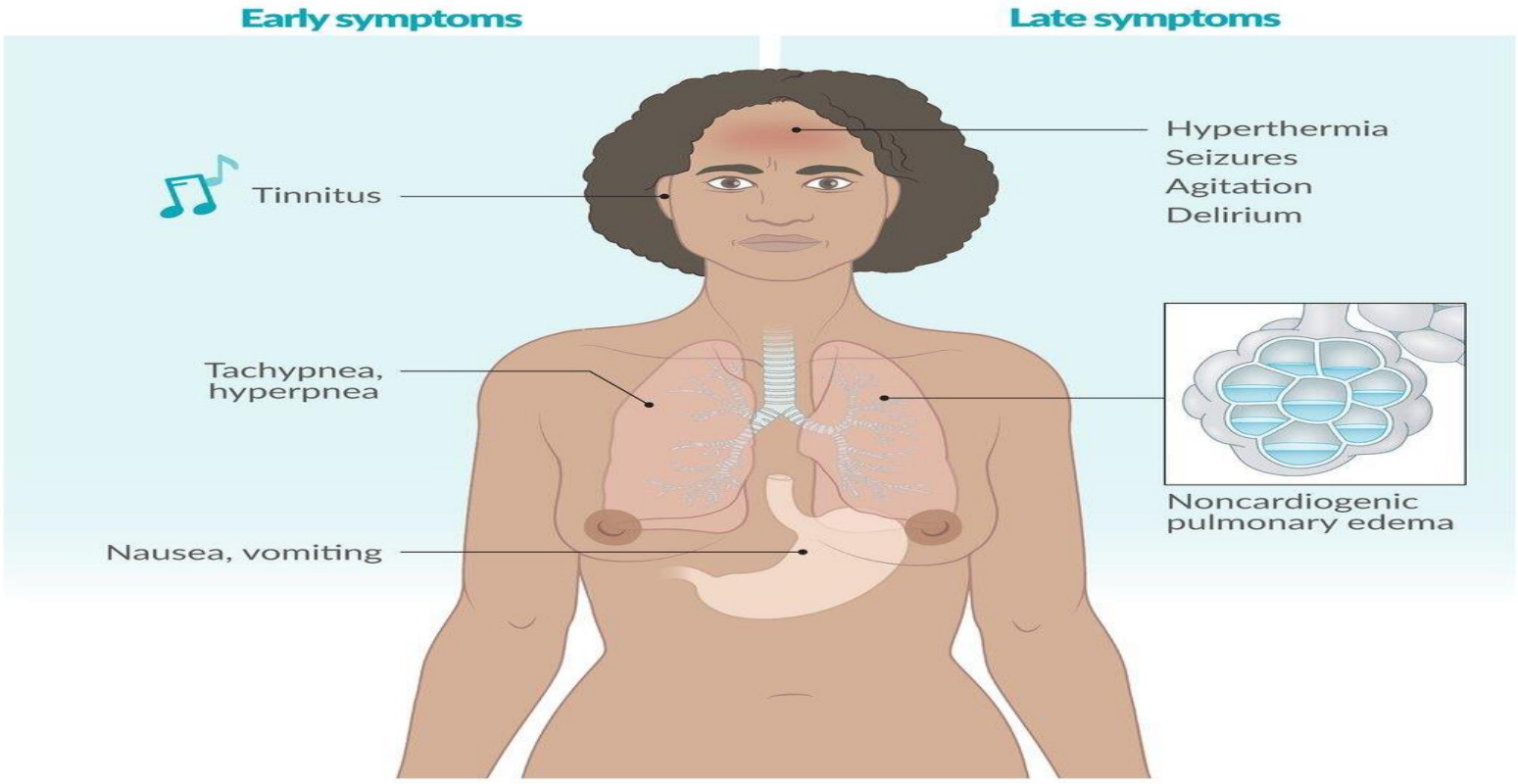
An anion gap metabolic acidosis and a respiratory alkalosis.

Classic presentation of salicylate toxicity

Salicylate poisoning

Diagnostics
- ABG: mixed respiratory alkalosis and increased anion gap metabolic acidosis
- Serum salicylate level: > 40 mg/dL
  BMP: hypokalemia, ↑ BUN, ↑ creatinine
- Serial serum salicylate level (every 2 hours) until downtrending
- Salicylate level may be falsely low within 4 hours of ingestion

Treatment
- Stabilization of vital signs
- Gastric decontamination: consider indications for oral activated charcoal or gastric lavage
- Fluid and acid-base management: fluid resuscitation, alkalinization with IV sodium bicarbonate
- Hemodialysis in severe cases



Case 2:

A 68-year-old man with a long history of smoking and COPD presents with worsening shortness of breath. ABG: pH 7.33, $PaCO_2$ 55 mmHg, HCO_3^- 30 mEq/L. Electrolytes: Na^+ 140, Cl^- 100, HCO_3^- 30

- A chronic, compensated respiratory acidosis
- Acute compensated respiratory acidosis
- Metabolic acidosis with high anion gap
- Metabolic acidosis with non ionic gap

Case 2:

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- Chronic, compensated respiratory acidosis
- Acute compensated respiratory acidosis
- Metabolic acidosis with high anion gap
- Metabolic acidosis with non ionic gap

Case 3:

A 35-year-old man with 3 days of persistent vomiting presents with dehydration. ABG: pH 7.50, $PaCO_2$ 48 mmHg, HCO_3^- 34 mEq/L Serum electrolytes: Na^+ 138, Cl^- 92, K^+ 3.2, HCO_3^- 34.

- **Chronic uncompensated metabolic acidosis**
- **Chronic uncompensated metabolic alkalosis**
- **Acute compensated metabolic acidosis**
- **Acute compensated metabolic alkalosis**
- **Acute uncompensated metabolic alkalosis**
- **Acute uncompensated metabolic acidosis**
- **Chronic compensated metabolic acidosis**
- **Chronic compensated metabolic alkalosis**

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- **A. Chronic uncompensated metabolic acidosis**
- **B. Chronic uncompensated metabolic alkalosis**
- **C. Acute compensated metabolic acidosis**
- **D. Acute compensated metabolic alkalosis**
- **E. Acute uncompensated metabolic alkalosis**
- **F. Acute uncompensated metabolic acidosis**
- **G. Chronic compensated metabolic acidosis**
- **H. Chronic compensated metabolic alkalosis**

Case 4:

A 70-year-old woman with sepsis presents with confusion and hypotension. Lactate is 5 mmol/L. ABG: pH 7.20, $PaCO_2$ 25 mmHg, HCO_3^- 10 mEq/L. Serum electrolytes: Na^+ 135 mEq/L, Cl^- 95 mEq/L, HCO_3^- 10 mEq/L.
Which of the following best describes her acid–base disorder?

- A. High Anion Gap Metabolic Acidosis with appropriate respiratory compensation.
- B. High Anion Gap Metabolic Acidosis without appropriate respiratory compensation.
- C. Normal Anion Gap Metabolic Acidosis with appropriate respiratory compensation.
- D. Normal Anion Gap Metabolic Acidosis without appropriate respiratory compensation.

Case 4:

A 70-year-old woman with sepsis presents with confusion and hypotension. Lactate is 5 mmol/L. ABG: pH 7.20, $PaCO_2$ 25 mmHg, HCO_3^- 10 mEq/L. Serum electrolytes: Na^+ 135 mEq/L, Cl^- 95 mEq/L, HCO_3^- 10 mEq/L.
Which of the following best describes her acid–base disorder?

- A. High Anion Gap Metabolic Acidosis with appropriate respiratory compensation.
- B. High Anion Gap Metabolic Acidosis without appropriate respiratory compensation.
- C. Normal Anion Gap Metabolic Acidosis with appropriate respiratory compensation.
- D. Normal Anion Gap Metabolic Acidosis without appropriate respiratory compensation.

Questions?



References & further reading

- Step-up to medicine 5th edition
- 1st Aid USMLE Step 1, 2020 edition
- Amboss
- OSMOSIS
- DAVIDSON 21st Edition

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Thank You